

AMIRKABIR UNIVERSITY OF TECHNOLOGY

Advanced Calculus and Algorithms

Saad Mollaei

Math and Computer Science Department

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1 Types of Convergences

in the table 1 six types of convergences are compared.

Type	Definition	Short Explanation
Pointwise	$f_n(x) \rightarrow f(x)$ for every x	Each point converges independently.
Uniform	$\sup_x f_n(x) - f(x) \rightarrow 0$	Same convergence rate on the whole domain.
Almost Everywhere	$f_n(x) \rightarrow f(x)$ except on a set of measure zero	Failure is allowed on negligible sets.
In Measure	$\mu(\{ f_n - f > \varepsilon\}) \rightarrow 0$	Large errors become increasingly rare.
L^p Convergence	$\ f_n - f\ _p \rightarrow 0$	Average error measured by the p -norm vanishes.
Weak Convergence	$\ell(f_n) \rightarrow \ell(f)$ for all continuous linear functionals ℓ	Convergence is tested through functionals.

Table 1: Comparison of Different Types of Convergence

2 Norm Problem

$$A \in M_{n \times n}(\mathbb{R})$$

$$\lambda \in \mathbb{R}$$

$$\|A\|_{op} < |\lambda|$$

We multiply both sides by the norm of an arbitrary element of $\mathbb{R}^n$.

$$\|A\|_{op} \|v\| < |\lambda| \|v\| \text{ for all nonzero } v \in \mathbb{R}^n$$

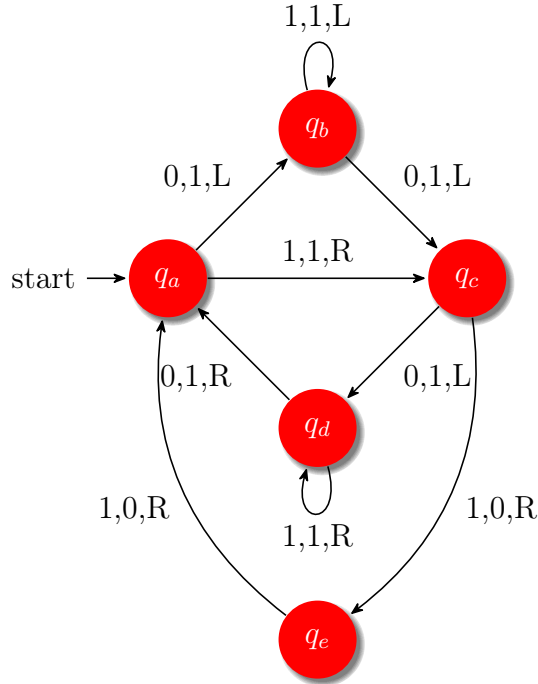
Knowing $\|Ax\| \leq \|A\|_{op} \|x\| \quad \forall x \in \mathbb{R}^n$ we get the following,

Thus, $A - \lambda I_n$ is invertible.

Because otherwise λ would be an eigenvalue for A ;

meaning, $\|Av\| = |\lambda| \|v\|$ for some nonzero $v \in \mathbb{R}^n$, contradicting line ($\star$).

2.1 Representing The Norm Problem Using Automata



The current candidate for the busy beaver for five states. It is presumed that this Turing machine writes a maximum number of 1's before halting among all Turing machines with five states and the tape alphabet $\{0, 1\}$. Proving this conjecture is an open research problem.

■

3 Calculus Concepts Illustrated

The following figure presents two fundamental ideas in calculus. Subfigure 1a shows the tangent line to the parabola $f(x) = x^2$ at $x = 1$, representing the derivative $f'(1) = 2$. Subfigure 1b illustrates the definite integral $\int_0^1 x^2 dx$ as the area under the same curve from $x = 0$ to $x = 1$.

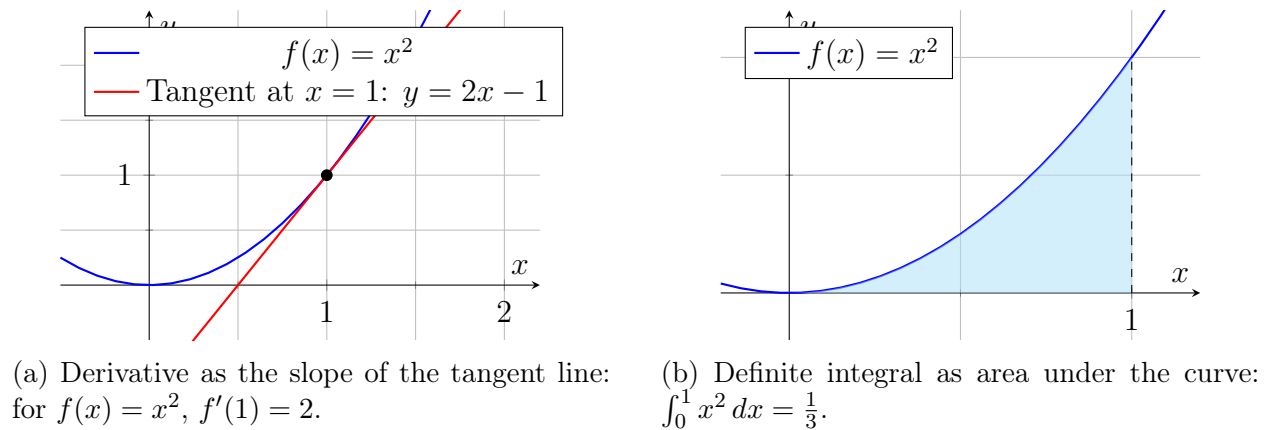


Figure 1: Calculus illustrated: derivative (left) and definite integral (right).

As seen in Figure 1a, the derivative gives the instantaneous rate of change; the slope of the tangent line at $(1, 1)$ is 2. Meanwhile, Figure 1b visualizes the accumulation of area under $f(x) = x^2$ from 0 to 1, which evaluates to $\frac{1}{3}$. Together, these subfigures capture the essence of differential and integral calculus.

4 Euclid's Algorithm GCD[3]

Euclid's algorithm computes the GCD of two positive integers. The algorithm works by repeatedly replacing the larger number with the remainder of dividing the larger by the smaller, until the remainder becomes zero. The last non-zero remainder is the GCD.

4.1 The Algorithm

Algorithm 1: Euclid's algorithm (remainder version)

Input : Two positive integers a and b

Output: $\gcd(a, b)$

```

1 while  $b \neq 0$  do
2    $r \leftarrow a \bmod b$  ;
3    $a \leftarrow b$  ;
4    $b \leftarrow r$  ;
5 end
6 return  $a$ 

```

4.2 Explanation

The algorithm proceeds as follows:

- The **loop condition** (line 1) checks whether b is zero. If $b = 0$, the algorithm exits and returns a as the GCD (line 6).
- Inside the loop, we compute the remainder $r = a \bmod b$ (line 2), then update a to the old value of b (line 3), and finally set b to the remainder r (line 4).
- This process reduces the pair (a, b) to $(b, a \bmod b)$, which preserves the GCD. The loop repeats until the remainder is zero.

For example, computing $\gcd(48, 18)$: $48 \bmod 18 = 12 \rightarrow$ new pair $(18, 12)$; $18 \bmod 12 = 6 \rightarrow$ new pair $(12, 6)$; $12 \bmod 6 = 0 \rightarrow$ new pair $(6, 0)$; loop ends, returns 6.

4.3 Flowchart of Euclid's Algorithm

Figure 2 shows the same logic as a flowchart, with each step corresponding directly to the numbered lines in Algorithm 1.

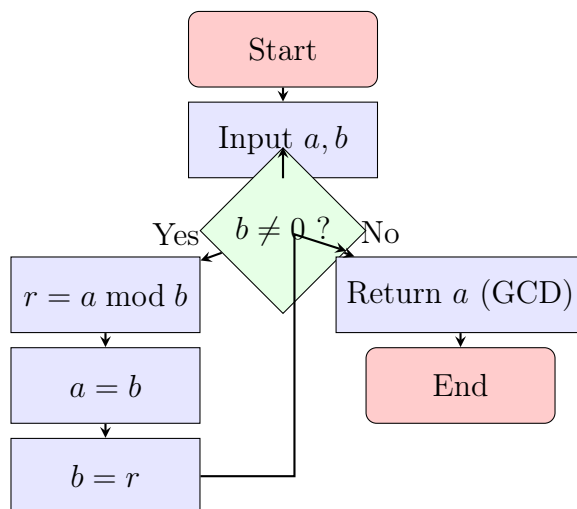


Figure 2: Flowchart of Euclid's algorithm. The loop (Yes branch) corresponds to lines 1 to 4 in Algorithm 1. The exit branch (No) returns the GCD as in line 6.

The flowchart makes the iteration explicit: after updating a and b , the control loops back to the condition $b \neq 0$, exactly as the `while` loop does in the algorithm.

5 Examining Unboundedness of The Space

$V = P[0, 1]$ is the space of polynomials from $[0, 1]$ to $\mathbb{R}$.

$T : V \rightarrow V \quad T(P) = P' \quad$ is the linear differentiation function.

And the norm used here is the supnorm.

$$A = \{x^n | n \in \mathbb{N}\} \subset V$$

$$\|x^n\| = 1 \quad \forall n \in \mathbb{N} \quad \text{thus } A \text{ is in the closed unit ball. } (\star)$$

$T(A) = \{nx^{n-1} | n \in \mathbb{N}\}$ is unbounded.

$$\because \forall M \in \mathbb{R} \quad \|(\lfloor M \rfloor + 1)x^{\lfloor M \rfloor}\| = \lfloor M \rfloor + 1 > M$$

Thus T is unbounded; in particular since $(\star)$ holds, we get that T is unbounded on the closed unit ball.

Now by Theorem 2.2.5 presented on page 116 of the book by Apostol[2], the continuity of T is disproved. ■

5.1 Discontinuity

We'll prove discontinuity for $p_0 = 0$ explicitly.

Writing the ϵ - δ definition for the continuity of $p_0 = 0$,

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall p \in V \quad \|p\| < \delta \rightarrow \|T(p)\| < \epsilon$$

negating it,

$$\exists \epsilon > 0 \quad \forall \delta > 0 \quad \exists p \in V \quad \|p\| < \delta \quad \wedge \quad \|T(p)\| \geq \epsilon$$

Choose $\epsilon = \frac{1}{2}$, $p = \frac{x^{n+1}}{n+1}$ taking n so big that $\frac{1}{n+1} < \delta$

substituting, $\|p\| = \frac{1}{n+1} < \delta$ and $\|T(p)\| = \|x^n\| = 1 \geq \frac{1}{2} = \epsilon$ ■

Now the sequential definition.

$\left\{\frac{x^{n+1}}{n+1}\right\}$ uniformly converges to 0.

$\left\{T\left(\frac{x^{n+1}}{n+1}\right)\right\} = \{x^n\}$ does not even converge.

$\therefore$ (informal) For any large n , you can get x^n as close as you want to 1. ■

5.2 Parallelogram Law

Any norm induced by an inner product satisfies the Parallelogram law[3].

Proof.

$$\begin{aligned}
 \|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle && \text{Definition of Inner Product Norm} \\
 &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\
 &\quad + \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle && \text{Bilinearity of Inner Product} \\
 &= 2\langle x, x \rangle + 2\langle y, y \rangle \\
 &= 2(\|x\|^2 + \|y\|^2) && \text{Definition of Inner Product Norm}
 \end{aligned}$$

$$\|x_n\| \leq 1, \|y_n\| \leq 1, \lim_{n \rightarrow \infty} \|x_n + y_n\| = 2$$

By applying the two inequalities to $\|x_n + y_n\|^2 + \|x_n - y_n\|^2 = 2(\|x_n\|^2 + \|y_n\|^2)$,

we get that, $\|x_n + y_n\|^2 + \|x_n - y_n\|^2 \leq 4$

now onto the problem,

$$0 \leq \|x_n - y_n\|^2 \leq (4 - \|x_n + y_n\|^2)$$

$$0 \leq \|x_n - y_n\| \leq \sqrt{4 - \|x_n + y_n\|^2}$$

$$0 \leq \lim_{n \rightarrow \infty} \|x_n - y_n\| \leq \lim_{n \rightarrow \infty} \sqrt{4 - \|x_n + y_n\|^2} \quad \because \text{The Sandwich Theorem}$$

$$\lim_{n \rightarrow \infty} \sqrt{4 - \|x_n + y_n\|^2} = \sqrt{4 - \left(\lim_{n \rightarrow \infty} \|x_n + y_n\|\right)^2} = 0 \quad \because 0 \leq \|x_n + y_n\| \leq \|x_n\| + \|y_n\| \leq 2$$

$$\therefore \lim_{n \rightarrow \infty} \|x_n - y_n\| = 0$$

■

5.3 The Dual Norm

Definition of the "Dual norm"[2] with respect to the normed space $(V, \|\cdot\|)$:

$$f \in V^* \quad \|f\|^* = \max \{ |f(x)| : \|x\| = 1 \}$$

Take the normed space to be $(\mathbb{R}^n, \|\cdot\|_1)$,

$$\|(x_1, x_2, \dots, x_n)\|_1 = \sum_{i=1}^n |x_i|,$$

$e_1, e_2, \dots, e_n$ the standard basis

and $\phi_1, \phi_2, \dots, \phi_n$ the standard dual basis.

$$f \in (\mathbb{R}^n)^* = \mathcal{L}(\mathbb{R}^n, \mathbb{R}) \quad \text{so, } f = \sum_{i=1}^n a_i \phi_i,$$

$$\begin{aligned} \|f\|_1^* &= \max \left\{ \left| \sum_{i=1}^n a_i \phi_i(x) \right| : \|x\|_1 = 1 \right\} \\ &= \max \left\{ \left| \sum_{i=1}^n a_i x_i \right| : \sum_{i=1}^n |x_i| = 1 \right\}, \end{aligned}$$

$$\left| \sum_{i=1}^n a_i x_i \right| \leq \sum_{i=1}^n |a_i x_i| \leq \sum_{i=1}^n |a_i| |x_i| \leq \sum_{i=1}^n M |x_i| = M,$$

$$\begin{aligned} \text{where } M &= \max \{ |a_i| : i \in \{1, 2, \dots, n\} \} \\ &= |a_k| \quad \text{for some } k \in \{1, 2, \dots, n\}. \end{aligned}$$

Now by choosing $x_k = 1$ and all other x 's to be 0, we can get as high as M .

thus, $\|f\|_1^* = M = |a_k| = \|(a_1, a_2, \dots, a_n)\|_{\max}$

■

5.3.1 Exponential Case

We want to show that there does not exist a real matrix $A \in \mathbb{R}^{2 \times 2}$ such that its exponential is:

$$e^A = \begin{pmatrix} -1 & 0 \\ 0 & -4 \end{pmatrix}$$

5.3.2 Proof of the Theorem

Theorem: if λ is an eigenvalue of A , e^λ is an eigenvalue of e^A ; this accounts for all the eigenvalues of e^A .

So take λ_1 and λ_2 to be the eigenvalues of A , then the eigenvalues of e^A are e^{λ_1} and e^{λ_2} . Therefore, we require:

$$e^{\lambda_1} = -1 \quad \text{and} \quad e^{\lambda_2} = -4$$

Thus:

$$\begin{aligned} \lambda_1 &= (2k+1)\pi i \quad \text{for } k \in \mathbb{Z} \\ \lambda_2 &= 2\log 2 + (2k+1)\pi i \quad \text{for } k \in \mathbb{Z} \end{aligned}$$

For any real matrix A , the eigenvalues must either be real or come in complex conjugate pairs. However, this is not the case. ■

References

- [1] James Stewart, *Calculus: Early Transcendentals*, 8th ed., Cengage Learning, 2016.
- [2] Tom M. Apostol, *Calculus, Volume 1: One-Variable Calculus with an Introduction to Linear Algebra*, 2nd ed., Wiley, 1967.
- [3] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.